Current ROLE OVERVIEW
Position: Software Engineer, Temp
Location: Stanford School of Medicine Genomic Lab, Stanford, CA
Timeline: September 2025 - Present
Focus: Building full-stack AI/ML platform for regulatory genomics research

Work product PLATFORM ARCHITECTURE HIGHLIGHTS
Distributed Cloud Orchestration
The platform leverages a sophisticated distributed architecture rather than simple sequential API calls. Built on Google Cloud Platform (GCP) with Prefect for workflow orchestration and Kubernetes on Nautilus for compute execution, the system handles massive parallel processing of genomic variants.

Key architectural features:

Event-driven architecture using Pub/Sub for job queuing and decoupling services

Dynamic resource allocation across 100+ CPU nodes and 5 A100 GPU nodes

Asynchronous processing with async_gather() for parallel job coordination

Multi-cloud capabilities using Terraform for infrastructure as code

Fault-tolerant pipelines with automatic retries and completion workflows

Intelligent Job Scheduling
Instead of blocking sequential calls, the platform implements priority-based job scheduling:

Priority 3: Annotation jobs (foundational genomic annotations)

Priority 4: AlphaGenome multimodal scoring (optional, when enabled)

Priority 5: ChromBPNet per-model scoring (highest parallelism, runs across multiple cell types simultaneously)

Jobs are pulled from Pub/Sub based on available K8s capacity and routed to appropriate GPU pools, enabling efficient resource utilization.

CORE ML MODEL INTEGRATIONS
ChromBPNet Integration
The platform integrates ChromBPNet (Pampari et al., 2024), a bias-factorized deep learning model for base-resolution chromatin accessibility prediction.

What it does:

Predicts chromatin accessibility profiles from DNA sequence

Deconvolves Tn5/DNase-I enzyme bias from true regulatory signals

Identifies transcription factor footprints at single-base resolution

Discovers predictive motif lexicons and cooperative syntax

Platform implementation:

Users can train custom ChromBPNet models for any cell type

Pre-trained models available for ENCODE cell lines (K562, GM12878, HepG2, H1-hESC, IMR90)

Automated bias correction using chromatin-background bias models

TF-MoDISco integration for de novo motif discovery from contribution scores

Variant scoring outputs:

Log fold-change (logFC): Change in total accessibility magnitude

Jensen-Shannon Distance (JSD): Change in base-resolution profile shape

Active Allele Quantile (AAQ): Calibration against genome-wide accessibility

Integrative Effect Size (IES): Product of logFC and JSD

Integrative Prioritization Score (IPS): Product of logFC, JSD, and AAQ

AlphaGenome Integration
The platform leverages AlphaGenome (Avsec et al., 2026), a unified multimodal sequence-to-function model from Google DeepMind.

What it does:

Takes 1Mb DNA sequence as input

Predicts 5,930 human or 1,128 mouse functional genomics tracks

Covers 11 modalities: gene expression, splicing, chromatin accessibility, histone modifications, TF binding, contact maps

Achieves base-pair resolution predictions with long-range context

Platform implementation:

Uses the distilled student model for efficient inference

Parallel scoring across multiple modalities in a single API call

Tissue-specific effect predictions (49 GTEx tissues)

Splice site/junction disruption analysis

Key capabilities leveraged:

eQTL effect prediction (Spearman ρ 0.49 vs 0.39 for Borzoi)

sQTL classification (auPRC 0.80 vs 0.75)

Chromatin accessibility QTL prediction across ancestries

Enhancer-gene linking without explicit training

```
User Upload (VCF/TSV) → GCP Storage → Pub/Sub Job Queue
   ↓
PARALLEL EXECUTION (Distributed, not sequential):
   ├── Annotation Layer (Priority 3):
   │   ├── Ensembl VEP for gene annotations
   │   ├── Population frequencies (gnomAD/dbSNP)
   │   ├── Clinical significance (ClinVar)
   │   ├── Trait associations (GWAS Catalog)
   │   └── Regulatory element overlaps (ENCODE)
   │
   ├── ChromBPNet Scoring (Priority 5):
   │   ├── Runs independently for EACH selected cell type model
   │   ├── Parallel GPU execution across models
   │   ├── Base-resolution accessibility prediction
   │   ├── Contribution scoring with DeepLIFT
   │   └── TF motif impact analysis
   │
   └── AlphaGenome Scoring (Priority 4, optional):
       ├── Multimodal prediction in single pass
       ├── RNA-seq expression effects
       ├── Splice site/junction disruption
       ├── Chromatin accessibility changes
       ├── TF binding affinity changes
       └── Contact map disruption
   ↓
async_gather() - Wait for ALL parallel jobs
   ↓
```

```
BigQuery JOIN (Materialize all results)
   ↓
GCP Storage Outputs:
   ├── variant_summary.tsv.gz (complete dataset)
   └── variant_summary_prioritized.tsv.gz (high-confidence)
   ↓
Knowledge Graph Construction (Neo4j)
   ↓
LLM Analysis & Visualization
```

LLM AGENT & KNOWLEDGE GRAPH LAYER
Knowledge Graph Construction
After materialization, results are transformed into a Neo4j knowledge graph:

Node types:

Variants: Genomic coordinates, alleles, all prediction scores

Genes: Annotations, pathways, disease associations

Transcription Factors: Motifs, cell-type specificity, ChIP-seq evidence

Cell Types/Tissues: Accessibility profiles, model availability

Regulatory Elements: Enhancers, promoters, insulators

Biological Pathways: GO terms, KEGG, Reactome

Edge relationships:

variant→gene (regulatory impact scores, distance, direction)

variant→TF (motif disruption, binding affinity changes)

TF→gene (regulation evidence, ChIP-seq peaks)

variant→cell_type (effect specificity across contexts)

gene→pathway (functional annotations)

LLM Multi-Agent Architecture
The platform implements a sophisticated agent-based LLM system:

```
User Query
   ↓
```

```
Query Classifier Agent
    ├── Technical Architecture → Orchestration & Infrastructure Specialist
    ├── ML Model Questions → ChromBPNet/AlphaGenome Expert
    ├── Biological Interpretation → Genomics Domain Scientist
    ├── Variant Analysis → Regulatory Effects Specialist
    └── Visualization Requests → Data Visualization Agent
  ↓
Context Retrieval (Pinecone + Knowledge Graph)
  ↓
Response Generation with:
    ├── Cited evidence from papers
    ├── Confidence scores for predictions
    ├── Visual explanations (plots, tracks)
    └── Follow-up probing suggestions
```

LLM Analysis Capabilities:

Mechanism explanation: "This variant disrupts a GATA motif in an erythroid enhancer, reducing accessibility by 40%"

Cross-modal reasoning: Connecting accessibility changes to expression effects

Literature integration: Linking findings to published studies

Hypothesis generation: Suggesting follow-up experiments

Comparative analysis: Contrasting effects across cell types

TECHNICAL SKILLS DEMONSTRATED

Cloud & Infrastructure

Google Cloud Platform: Compute Engine, Cloud Composer, Pub/Sub, BigQuery, Cloud Storage

Kubernetes: Managing GPU/CPU pools on Nautilus cluster, job scheduling, resource allocation

Terraform: Infrastructure as code for multi-cloud deployments

Docker: Containerization of ML models and analysis pipelines

Prefect: Workflow orchestration with priority-based job queuing

Backend & APIs

FastAPI: REST API development for platform services

Pub/Sub: Event-driven architecture for decoupled services

Async Python: async_gather() for parallel job coordination

BigQuery: Large-scale genomic data joins and materialization

Neo4j: Knowledge graph construction and querying

ML & Data Science
ChromBPNet: Bias-factorized accessibility modeling, TF-MoDISco, contribution scoring

AlphaGenome: Multimodal prediction, variant effect scoring, ISM analysis

PyTorch: Model serving and inference optimization

DeepLIFT/DeepSHAP: Model interpretability for genomic sequences

Genomic tools: PyBigWig, pyranges, VEP, Ensembl REST API

Frontend & Visualization
React/TypeScript: User interface for variant upload and analysis

D3.js/GenomeD3: Genomic track visualizations

Cytoscape.js: Knowledge graph exploration

WebSocket: Real-time analysis updates

DevOps & Monitoring
Prometheus/Grafana: System monitoring and alerting

GitHub Actions: CI/CD for model and code deployment

Vercel/Render: Frontend hosting and deployment

KEY ACHIEVEMENTS & IMPACT
Performance Improvements
Reduced pipeline failure rates from ~20% to 1-2% through improved validation logic, error handling, and modular pipeline design

Parallelized variant scoring achieving 1000+ variants per minute

Optimized GPU utilization across 5 A100 nodes with intelligent job scheduling

Technical Innovations
Built the first platform integrating both ChromBPNet and AlphaGenome for comprehensive regulatory analysis

Implemented priority-based job queuing that dynamically allocates resources based on workload

Created a knowledge graph linking ML predictions with external biological databases

Developed an LLM agent system specialized for genomic variant interpretation

Research Enablement
Platform enables researchers to:

Validate GWAS hits in disease-relevant cell types

Interpret de novo variants in rare disease patients

Study TF cooperativity through composite motif analysis

Prioritize drug targets based on regulatory impact

Cross-species comparison of regulatory elements